MA2202

Fermat's little theorem

Let p be a prime number. Then
$$p|(n^p - n) \quad \forall n \in \mathbb{Z}$$
i.e.
$$n^p \equiv n \pmod p \implies n^{p-1} \equiv 1 \pmod p$$
Applying this idea,
$$n^{a(p-1)+b} \equiv n^b \pmod p$$

Group isomorphisms

Definition

If $G \cong H$:

- If G is abelian, H is abelian.
- $\phi : G \to G$ given by $\phi(g) = g^{-1}$ is a group isomorphism $\iff$ G is an abelian group.

Two isomorphisms

Suppose $\phi : (G, *) \to (H, \star)$ and $\psi : (H, \star) \to (K, \cdot)$ are two isomorphisms of groups. Then

- the inverse function $\phi^{-1} : (H, \star) \to (G, *)$ and
- the composite function $\psi \circ \phi : (G, *) \to (K, \cdot)$

are group isomorphisms.

Subgroups

Roots of unity $(\mu_m, \times)$ is a subgroup of $(\mu_n, \times)$ if $m|n$.

- Let $(G, *)$ be a group and let $H \subseteq G$ be a nonempty subset. Then $(H, *)$ is a subgroup iff for all $a, b \in H$, we have $a * b^{-1} \in H$.
- Intersection of a finite collection of subgroups is a non-empty subgroup.
- Let $(H, *)$ and $(K, *)$ be subgroups of $(G, *)$. If $(H \cup K, *)$ is a subgroup, then either $H \subseteq K$ or $K \subseteq H$.

Cyclic notations

Fix $f \in S_n$. Let $x \in X = \{1, \cdots, n\}$. Consider the sequence of integers in X: $x_0, x_1, x_2, \cdots$, where $x_0 = x$ and $x_i = f^i(x) \in X$.

- Since X is finite, the sequence will repeat. Let x_r be the first integer that repeats in the sequence. Can be shown that $x_r = x_0 = x$.
- $\mathcal{O} = \{x_0, x_1, \cdots, x_{r-1}\}$ is an orbit of the powers of f.
- The sequence $(x_0 x_1 \cdots x_{r-1})$ is called a cycle.
- $X = \bigsqcup_j \mathcal{O}_j$

Example $f = (16)(24)(3789)(5)$

- f is also equal to $(61)(24)(8937)(5)$. We can rotate within the cycle.
- f is also equal to $(16)(24)(3789)$. We can drop singleton cycles.
- $h = (16)$ is the bijection in S_9 such that $h(1) = 6, h(6) = 1$ and $h(x) = x$ for $x \neq 1, 6$.
- f is also equal to $(24)(16)(3789)(5)$. We can swap the cycles because they represent bijections in S_9 which are disjointed cycles and they are commutative.

Cyclic permutation A bijection $h \in S_n$ which is represented by a single cycle is called a cyclic permutation or cycle. Two cycles
$$h = (i_1 \cdots i_r) \text{ and } h' = (j_1 \cdots j_s)$$
are called disjointed cycles if $i_\alpha \neq j_\beta$ for all $\alpha = 1, \cdots, r$ and $\beta = 1, \cdots, s$.

Theorem 23 Let $f \in S_n$. Then

- $f = h_1 \circ h_2 \circ \cdots \circ h_r$ can be factorized into a product of mutually disjointed cycles.
- The factorization is unique up to an ordering of the product of cycles, i.e. if
$$f = h_1 \circ h_2 \circ \cdots \circ h_r = k_1 \circ k_2 \circ \cdots \circ k_s$$
are two factorization into mutually disjointed cycles, then by renaming the cycles k_i if necessary, we have $r = s$ and $h_i = k_i$ for $i = 1, \cdots, r$.

Transpositions A cycle $h \in S_n$ of the form $h = (ij)$ is a transposition.

- $(i_1 i_2 \cdots i_r) = (i_1 i_r)(i_1 i_{r-1}) \cdots (i_1 i_2)$. Hence, a cycle is a product of transpositions.
- Since $f \in S_n$ is a product of cycles, f is also a product of transpositions.

The sign character

Lemma For all permutation matrices $F, H \in S_n''$,

- $\det(F) = \det(F^T) = \pm 1$.
- $\det(FH) = \det(F)\det(H)$.

Proposition 25 Let $P(\boldsymbol{x}) = P(x_1, \cdots, x_n) = \prod_{1 \le i < j \le n} (x_i - x_j)$. For $f \in S_n$, let
$$P_f(\boldsymbol{x}) = P_f(x_1, \cdots, x_n) = P(x_{f(1)}, \cdots, x_{f(n)}) = \prod_{1 \le i < j \le n} (x_{f(i)} - x_{f(j)})$$

- $P_f(\boldsymbol{x}) = P(\boldsymbol{x})$ or $-P(\boldsymbol{x})$. We write $P_f(\boldsymbol{x}) = \operatorname{sgn}(f)P(\boldsymbol{x})$, where $\operatorname{sgn}(f) = \pm 1$.
- $\operatorname{sgn}(f \circ h) = \operatorname{sgn}(f)\operatorname{sgn}(h)$.

Even/odd Let $f, h \in S_n$.

- f is an even permutation if $\operatorname{sgn}(f) = 1$, and odd if $\operatorname{sgn}(f) = -1$.
- If f and h are both even (odd), then $f \circ h$ is even (odd).
- If f is odd and h is even, then $f \circ h$ is odd.
- A transposition is an odd permutation.
- A product of an even (odd) number of transpositions is even (odd).
- f is even $\iff$ f is a product of an even number of transpositions.

Alternating group Let
$$A_n = \{f \in S_n : \operatorname{sgn}(f) = 1\} = \{f \in S_n : f \text{ even}\}$$
be the set of all even permutations in S_n. Then $(A_n, \circ)$ is a subgroup of $(S_n, \circ)$.

- The subset of odd permutations is not a subgroup.

Cayley's theorem

Let $(G, *)$ be a finite group of order n. Then $(G, *)$ is isomorphic to a subgroup of $(S_n, \circ)$.

Proof

- We know that $(S_Y, \circ)$ is isomorphic to $(S_n, \circ)$.
- Let $Y = G$. For every $g \in G$, define function $f_g : Y \to Y$ by
$$f_g(y) = g * y \text{ for all } Y = G$$
Then construct $\phi : G \to S_Y$ by $\phi(g) = f_g$. ϕ is an injective group homomorphism, so G is isomorphic to the image G' which is a subset of S_Y, i.e. G is isomorphic to a subgroup of $(S_Y, \circ)$.

Cosets and Lagrange's theorem

Coset

Let H be a subgroup of G. For $g \in G$, denote
$$gH = \{gh : h \in H\} \text{ and } Hg = \{hg : h \in H\}$$
These are called a left coset and a right coset of H in G respectively. Note that $eH = He = H$.

- If G is abelian, then a left coset is also a right coset.

Mutually disjointed subsets

Let S be a set, and let $\{S_i : i \in I\}$ be a collection of subsets of S.

- We say that $\{S_i : i \in I\}$ is a collection of mutually disjointed subsets if $S_i \cap S_j = \emptyset$ for every distinct $i, j \in I$.
- We say that $\{S_i : i \in I\}$ forms a partition of S if it is a collection of mutually disjointed subsets, and $S = \bigcup_{i \in I} S_i$. We write $S = \coprod_{i \in I} S_i$.

Proposition 37

Let G be a group and let H be a subgroup. Let $x, y, z \in G$.

i. If $z \in xH$, then $zH = xH$.
ii. If $xH \cap yH \neq \emptyset$, then $xH = yH$.
iii. The collection of left cosets $\{xH : x \in G\}$ forms a partition of G.
iv. Every coset xH is of the same cardinality as H, i.e. there is a bijection $f : H \to xH$. If H is a finite group, then $|H| = |xH|$.

Definition

- Denote $G/H = \{xH : x \in G\}$ and $H \backslash G = \{Hx : x \in G\}$.
- Let $[G : H]$ denote the number of distinct left cosets of H in G, i.e. $[G : H] = |G/H|$. It is called the index of H in G.

Lagrange's Theorem

Let G be a finite group and let H be a subgroup.

- $|H|$ divides $|G|$.
- $[G : H] = |G/H| = |G| / |H|$.
- $[H : G] = |H \backslash G| = |G| / |H|$.

Corollary Let p be a prime integer, and let G be a group of order p.

- The only subgroups of G are $\{e\}$ or G.
- Let $x \in G$ and $x \neq e$. Let $x = \langle x \rangle = \{x^n : n \in \mathbb{Z}\}$ be the cyclic subgroup of G generated by x. Then $G = \langle x \rangle$.

Corollary If H is a subgroup of G and K is a subgroup of H, then
$$[G : K] = [G : H][H : K]$$

Generators

Subgroup Let G be a group, and let $X \subseteq G$. Let $S = \{H : H \text{ subgroup of } G, H \supseteq X\}$. We define
$$\langle X \rangle = \bigcap_{H \in S} H$$
and we call $\langle X \rangle$ the subgroup of G generated by X.

- If H is a subgroup of G containing X, then by definition, H contains $\langle X \rangle$. Hence, $\langle X \rangle$ is also called the smallest subgroup of G containing X.
- If the subgroup $\langle X \rangle = G$, then we say that G is generated by X.
- We say that a group G is finitely generated if it is generated by some finite subset. G could still be infinite, e.g. $G = (\mathbb{Z}, +)$ is generated by $X = \{1\}$.

Word A word on X is either e or a finite product $x_1^{r_1} x_2^{r_2} \cdots x_n^{r_n} \in G$ where $x_i \in X$ and $r_i \in \mathbb{Z}$ for $i = 1, \cdots, n$.

- Some x_i can be the same.
- Some r_i may be negative integers.
- If G is non-abelian, order of multiplication matters.
- Two different words may represent the same element in G.

Proposition 39

Let X be a subset of a group G. Let W be the set of words on X. Then W is a subgroup and $W = \langle X \rangle$.

Cyclic groups

Proposition Let $(G, *)$ be a group. Pick $a \in G$. The subset $\langle a \rangle = \{a^n \in G : n \in \mathbb{Z}\}$ is a subgroup of $(G, *)$. It is called the cyclic subgroup of G generated by a.

- $\langle a \rangle = \langle a^{-1} \rangle$.

Proposition 40 The order of the subgroup $|\langle a \rangle|$ is equal to the order $\mathrm{o}(a)$.

Proposition 41 Let G be a finite group. Let $a \in G$. Then $\mathrm{o}(a)$ divides $|G|$.

Corollary 42 Let G be a finite group of order p where p is a prime number. Pick $a \in G$ and $a \neq e$. Then
$$G = \langle a \rangle = \{e, a, \cdots, a^{p-1}\}$$

Cyclic group

Let $(G, *)$ be a group and let $x \in G$. A group $(G, *)$ is called a cyclic gp if $G = \langle x \rangle$ for some $x \in G$, i.e.
$$G = \langle x \rangle = \{x^n \in G : n \in \mathbb{Z}\}$$

- Group G is cyclic $\implies$ some element $x \in G$ has order $|G|$

Group homomorphisms

Let $(G,*)$ and $(H,\star)$ be two groups. A function $\phi: G \to H$ is called a group homomorphism if

$$\phi(x * y) = \phi(x) \star \phi(y)$$

for all $x, y \in G$.

- There is no requirement on ϕ to be injective or surjective. But if ϕ is a bijection, then we have a group isomorphism instead.
- Composition of group homomorphisms is a group homomorphism.
- Let $\phi: (G,*) \to (H,\star)$ be an injective group homomorphism. Then $(G,*)$ is isomorphic to its image which is a subgroup of $(H,\star)$.

Proposition 43 Let $\phi: (G,*) \to (H,\star)$ be a group homomorphism.

i. Let e_G and e_H be identity elements of the groups G and H respectively. Then $\phi(e_G) = e_H$.
ii. For all $g \in G$, $\phi(g^{-1}) = (\phi(g))^{-1}$.
iii. Let G' be a subgroup of G. Then the image $\phi(G')$ is a subgroup of H.
iv. Let H' be a subgroup of H. Then $\phi^{-1}(H')$ is a subgroup of G.

Kernel Let $\phi: (G,*) \to (H,\star)$ be a group homomorphism. The kernel of ϕ is defined as

$$\ker \phi = \phi^{-1}(e_H) = \{g \in G : \phi(g) = e_H\}$$

It is the set of elements in G that is sent to e_H under the mapping ϕ.

Prop. 44 Let $\phi: (G,*) \to (H,\star)$ be a group homomorphism and let K be the kernel of ϕ.

i. The kernel K is a subgroup of G.
ii. $\forall g_0 \in K$ and $g \in G$, we have $gg_0g^{-1} \in K$.
iii. For $g_0 \in G$, we have
$$\{g \in G : \phi(g) = \phi(g_0)\} = g_0K = Kg_0$$
i.e. every left coset of K is also a right coset.

Corollary 45 Let $\phi: (G,*) \to (H,\star)$ be a group homomorphism. Then ϕ is injective (as a function) $\iff \ker\phi = \{e_G\}$.

Group homomorphisms and subgps

Let $(G,*)$ and $(H,\star)$ be two groups and let $\phi: (G,*) \to (H,\star)$ be a group homomorphism. Let $K = \ker\phi$. Define

- $\mathbf{Sub}(G,K) = \{G' : G' \text{ subgroup of } G, G' \supset K\}$ which contains all the subgroups of G which contain K and
- $\mathbf{Sub}(H) = \{H' : H' \text{ subgroup of } H\}$

Define a function $\Phi: \mathbf{Sub}(G,K) \to \mathbf{Sub}(H)$ by $\Phi(G') = \phi(G')$ where $G' \in \mathbf{Sub}(G,K)$. By proposition 43(iii), $\phi(G')$ is a subgroup of H, so $\Phi(G') \in \mathbf{Sub}(H)$.

Theorem 46 Suppose ϕ is a surjective homomorphism. Then Φ is a bijection.

Normal subgroups

Let G be a group and let N be a subgroup.

- N is called a normal subgroup of G if for all $n \in N$ and $g \in G$, $gng^{-1} \in N$.
- We denote a normal subgroup N of G by $N \lhd G$.
- Suppose G is abelian. Then every subgroup N of G is a normal subgroup.

Prop. 48 The kernel of a group homomorphism $\phi: (G,*) \to (H,\star)$ is a normal subgroup of G.

Center Let $(G,*)$ be a group. Let

$$Z = \{z \in G : zg = gz \text{ for all } g \in G\}$$

Z is a normal subgroup of G and it is called the center of G.

Proposition 49 Let K be a subgroup of G. The following statements are equivalent.

i. The subgroup K is normal, i.e. for all $k \in K$ and $g \in G$, $gkg^{-1} \in K$.
ii. For all $g \in G$, $gKg^{-1} = K$.
iii. For all $g \in G$, $gK = Kg$, i.e. every left coset is also a right coset.
iv. For all $g \in G$, $(gK)(g'K) = (gg')K$.

Notation If K is a subgroup of G and $gK = g'K$ for some $g, g' \in G$, we write

$$g \equiv g' \pmod{_L K}$$

The subscript L in mod_L denotes left cosets.

Simple groups

A group G is simple if its normal subgroups are only its trivial normal subgroups $\{e\}$ and G.

- Let p be a prime number. $\mathbb{Z}/p\mathbb{Z}$ is a simple group.

Theorem 50 Let

$$A_n = \{f \in S_n : \mathrm{sgn}(f) = 1\} = \{f \in S_n : f \text{ even}\}$$

be the set of all even permutations in S_n.

- $(A_n, \circ)$ is a subgroup of $(S_n, \circ)$.
- For $n \neq 4$, the alternating group A_n is a simple group.

Lemma 51 Let H be a normal subgroup of A_n where $n \geq 5$. If H contains a 3-cycle, $H = A_n$.

- H contains a 3-cycle $\implies$ H contains all the 3-cycles of A_n. Every even permutation is the product of 3-cycles. Hence $H = A_n$.

Definition Let $X_n = \{1, 2, \cdots, n\}$. Recall that A_n is the set of even permutations on X_n. We can identify A_{n-1} as a subgroup of A_n by

$$A_{n-1} = \{\sigma \in A_n : \sigma(n) = n\}$$

Lemma 52 Let H be a normal subgroup of a group A. For subgroup A' of A, $H \cap A'$ is a normal subgroup of A'.

Quotient groups

Let $(G,*)$ be a group and let K be a normal subgroup. By proposition 49(iv), for all $g_1, g_2 \in G$, define the binary operation

$$(g_1K) \diamond (g_2K) = (g_1g_2)K$$

for $g_1K, g_2K \in G/K$.

Theorem 56

i. The pair $(G/K, \diamond)$ forms a group. It is called the quotient group of G by K.
ii. The function $\pi: (G,\star) \to (G/K, \diamond)$ defined by $\pi(g) = gK$ for all $g \in G$ is a surjective group homomorphism. It is called the quotient map or quotient homomorphism.
iii. The kernel of π is K.

The First Isomorphism Theorem

In this section, $(G,*)$ and $(H,\star)$ are (possibly infinite) groups. Let $\phi: (G,*) \to (H,\star)$ be a surjective group homomorphism. Let K be the kernel of ϕ.

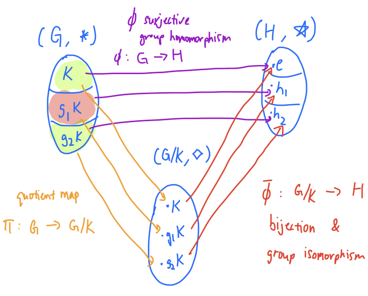
- Suppose $\phi(g) = h$ where $h \in H$ and $g \in G$. Then
$$\{x \in G : \phi(x) = h\} = gK$$
i.e. the whole of gK is sent to h under ϕ.

First Isomorphism Theorem

Let $\phi: (G,*) \to (H,\star)$ be a surjective group homomorphism. Let K be the kernel of ϕ. Then the function $\bar{\phi}: (G/K, \diamond) \to (H,\star)$ given by

$$\bar{\phi}(gK) = \phi(g)$$

is a well-defined group isomorphism.



- If ϕ is not surjective, then replace H with the image $H' = \phi(G)$ in the definition of $\bar{\phi}$.

Corollary Let $\phi: G \to H$ and $\psi: G \to H'$ be two group homomorphisms.

- Suppose ϕ and ψ have the same kernel K. Then, the images $\phi(G)$ and $\psi(G)$ are isomorphism groups.
- If G is a finite group, then
$$|\phi(G)| = |\psi(G)| = |G/K| = |G| \,/\, |K|$$

The Second Isomorphism Theorem

In this section, G is a group, M is a subgroup of G, and K is a normal subgroup of G.

Prop. 59 $MK = KM$ and it is a subgroup of G.

Proposition 60

i. The function $\phi: M \to MK/K$ defined by $\phi(m) = mK$ is a surjective group homomorphism.
ii. The kernel of ϕ is $M \cap K$. In particular, it is a normal subgroup of M.

Second Isomorphism Theorem

$$M/(M \cap K) \simeq (MK)/K$$

The Third Isomorphism Theorem

Let G be a group. Let M and K be normal subgroups of G such that $M \supseteq K$. Then M/K is a normal subgroup of G/K and

$$(G/K)/(M/K) \simeq G/M$$

If $M \not\supseteq K$, then replace K by $M \cap K$, which is a normal subgroup of G contained in M.

Corollary Let M and K be normal subgroups of G such that $M \supseteq K$. Then there is a surjective group homomorphism

$$\phi: G/K \to G/M$$

given by $\phi(gK) = gM$.

Euler's totient function

Let n be a positive integer. If $n = 1$, set $\Phi(1) = \{1\}$. Else set

$$\Phi(n) = \{x \in \mathbb{Z} : 0 \leq x \leq n, \gcd(x,n) = 1\}$$

- Let $*$ denote multiplication modulo n. Then $(\Phi(n), *)$ is a group.
- Let $\phi(n)$ denote the number of elements in $\Phi(n)$.
- For prime number p, $\Phi(p) = \{1, 2, \cdots p-1\}$, so $\phi(p) = p - 1$.
- For prime number p, let $n = p^r$.
$$\phi(p^r) = n - \frac{n}{p} = p^r\left(1 - \frac{1}{p}\right) = p^{r-1}(p-1)$$

Euler's theorem

Let x be an integer such that $\gcd(x,n) = 1$. Then

$$x^{\phi(n)} \equiv 1 \pmod{n}$$

Calculating $\phi(n)$

Suppose $n = p_1^{r_1}p_2^{r_2}\cdots p_n^{r_n}$. Then

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\cdots\left(1 - \frac{1}{p_k}\right) = \phi(p_1^{r_1})\phi(p_2^{r_2})\cdots\phi(p_k^{r_k})$$

Example Compute $43^{866} \pmod{360}$.

- $360 = 2^3 \cdot 3^2 \cdot 5$ so
$$\phi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 96$$
- Since $\gcd(43, 360) = 1$, Euler's theorem gives $43^{96} \equiv 1 \pmod{360}$.
- We have $866 = 96(9) + 2$ so
$$43^{866} \equiv 43^{96(9)+2} \equiv (43^{96})^9 43^2 \equiv 1^9 43^2 \equiv 49 \pmod{360}$$

Automorphism groups

Let $(G,*)$ be a group. An isomorphism $\phi: G \to G$ is called an automorphism of G. We denote the set of automorphisms of G by

$$\mathrm{Aut}(G) = \{\phi : G \to G : \phi \text{ is an isomorphism}\}$$

Isomorphism facts

- Identity map id_G is an isomorphism.
- Composition of isomorphisms is an isomorphism, i.e. $\circ$ is a binary operation on $\mathrm{Aut}(G)$.
- Inverse of an isomorphism is an isomorphism.

Proposition $(\mathrm{Aut}(G), \circ)$ forms a group.

- It is called the automorphism group of G.
- A subgroup A of $(\mathrm{Aut}(G), \circ)$ is called an automorphism subgroup.

Inner automorphism

Let G be a group and let $g \in G$. Then $\phi_g : G \to G$ given by
$$\phi_g(x) = gxg^{-1}$$
is a group automorphism. It is called an inner automorphism of G.

- Let $\mathrm{Inn}(G) = \{\phi_g : g \in G\}$ be the set of inner automorphisms.
- The subset $\mathrm{Inn}(G)$ is a normal subgroup of $(\mathrm{Aut}(G), \circ)$.

Proposition The map $T : G \to \mathrm{Inn}(G)$ given by $T(g) = \phi_g$ is a surjective group homomorphism whose kernel is the center of the group
$$Z(G) = \{z \in G : gz = zg \text{ for all } g \in G\}$$
By the first isomorphism theorem,
$$G/Z(G) \simeq \mathrm{Inn}(G)$$

The Sylow Theorems

Notation Let n be a positive integer. Suppose p^e divides n, but p^{e+1} does not divide n. We write $p^e || n$. Alternatively, n $= p^e m$ where $p \not| m$.

Definition

Let G be a finite group of order n. Let p be a prime divisor of n. Let H be a subgroup of order of p^e.

- H is called a p-subgroup of G.
- If $p^e || n$, then H is called a Sylow p-subgroup of G.

Example Let $G = S_9$. It has order $9! = 2^7 3^4 5^1 7^1$.

- A subgroup of order 2^5 is a 2-subgroup.
- A subgroup of order 2^7 is a Sylow 2-subgroup.

First Sylow Theorem

Let G be a group of order n. Let p be a prime divisor of n. Then G contains a Sylow p-subgroup.

Corollary Let G be a finite group of order n. Let p be a prime divisor of n. If $p^d | n$, then G contains a subgroup of order p^d.

Definition

Let P be a subgroup of G. Let $g \in G$. Then gPg^{-1} is a subgroup of G called a conjugate of P. Let P be a Sylow p-subgroup. Then a conjugate gPg^{-1} is also a Sylow p-subgroup.

Theorem 94

Let G be a group of order n. Let $\{P_1, P_2, \cdots, P_r\}$ be all the distinct conjugates of a Sylow p-subgroup $P = P_1$.

i. Let Q be a p-subgroup of G. Then $Q \subseteq P_i$ for some i.

ii. (Second) If Q is a Sylow p-subgroup of G, then $Q = P_i$ for some i.

iii. (Third) Let r denote the number of Sylow p-subgroups of G. Then
$$r \equiv 1 \pmod{p} \text{ and } r | [G : P]$$

Corollary 95 Let P be a Sylow p-subgroup of a finite group G. Then P is a normal subgroup $\iff$ P is the unique Sylow p-subgroup of G.